Pinoy Speak: A Trend Analysis and Contextual Lexicon System for Informal Filipino Text

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***Abstract*—Informal Filipino and Taglish evolve rapidly in online spaces, making them difficult to process using static dictionaries and conventional Natural Language Processing (NLP) resources. This paper presents Pinoy Speak, a deployed full-stack system for detecting, organizing, and explaining emerging Filipino internet slang. The system collects public text from selected Filipino Reddit communities, YouTube comments, and user chat interactions. As of the current implementation, it has archived 175,481 records and maintains approximately 104 verified slang entries. Candidate terms are processed through a pipeline that combines calamanCy tokenization, subword novelty signals, FastText nearest-neighbor evidence, temporal burstiness analysis, morphological filtering, and context-window disambiguation for ambiguous slang words. Verified candidates are enriched through a separate LLM pipeline, while the user-facing "Kuya Slang" tutor uses retrieval-augmented access to the live lexicon. Evaluation consisted of a 79-entry lexicon validation with three ratings per entry and a 36-response consented beta usability survey. Lexicon validation produced high definition correctness ratings (mean = 4.83/5) and example naturalness ratings (mean = 4.75/5), while beta usability results showed strong user acceptance across the dictionary, tutor, analyzer, navigation, and trend modules.**

***Index Terms*—Informal Filipino, Taglish, slang detection, lexical trends, natural language processing, FastText, calamanCy, retrieval-augmented generation.**

I. Introduction

THE Philippines has a highly active digital communication environment where informal language evolves rapidly across social media platforms, messaging applications, and online communities. In these spaces, Filipino users frequently combine Filipino, English, and localized internet expressions to form *Taglish*. This mixed-language practice is not random; previous linguistic work describes code-switching as a structured discourse strategy that can express identity, solidarity, emphasis, humor, and contextual meaning [1]–[3].

However, this same flexibility creates challenges for computational language processing. Many existing NLP tools are designed around formal, standardized, and high-resource language settings. Informal Filipino text often contains non-standard spelling, creative abbreviations, borrowed English words with localized meanings, syllable reversals, alphanumeric spellings, and context-dependent expressions. As a result, traditional dictionaries and static NLP resources may fail to recognize terms that are widely understood by Filipino internet users.

In this study, **slang** refers to informal or non-standard lexical items and expressions that are used within specific communities, often as markers of social identity or shared cultural knowledge. In Filipino digital discourse, slang may appear as a completely new word, a modified form of an existing word, or a familiar English or Filipino word that has acquired a new meaning in local online contexts. For example, terms such as *omsim* and *forda* are structurally novel, while words such as *bet*, *solid*, and *awit* require surrounding context to determine whether they are being used in their ordinary or slang sense.

A. Research Gap

Existing Filipino language resources are largely static and cannot easily reflect the pace of language change in online communities. A word may emerge, trend, shift meaning, and decline within a short period. Static dictionaries may eventually record some of these expressions, but they do not provide a mechanism for tracking when terms become popular, how they are used in context, or how meanings change across time.

Recent work on semantic change and diachronic word embeddings shows that changes in word meaning can be studied computationally by examining distributional patterns across time [4]–[6]. However, such methods are rarely applied to informal Filipino and Taglish data. In addition, Filipino internet slang often combines lexical novelty with code-switching, making it insufficient to rely only on dictionary lookup, word frequency, or direct translation.

B. Objectives

This study aims to develop and evaluate *Pinoy Speak*, a dynamic lexicon and trend-analysis system for informal Filipino text. Specifically, the study aims to:

- collect public informal Filipino and Taglish text from selected online sources;
- detect emerging slang candidates using lexical novelty, temporal burstiness, and semantic shift indicators;
- reduce false positives through morphological filtering and context-window disambiguation;
- enrich verified slang entries with definitions, examples, formation types, and usage context;
- provide user-facing tools such as a searchable dictionary, trend dashboard, concordance search, sentence analyzer, translator, and tutoring chatbot; and
- evaluate the usability and perceived usefulness of the deployed web application through beta user testing.

C. Scope and Delimitations

The study focuses on public informal Filipino and Taglish text from online sources where slang usage is expected to be frequent. The implemented Reddit collector targets 15 high-slang-density communities: *r/peyups*, *r/ADMU*, *r/dlsu*, *r/Tomasino*, *r/RateUPProfs*, *r/studentsph*, *r/CasualPH*, *r/OffMyChestPH*, *r/mobilelegendsPINAS*, *r/phgaming*, *r/OPM*, *r/PinoyProgrammer*, *r/ChikaPH*, *r/TiktokPH*, and *r/BPOinPH*. These communities were selected because they represent student life, university discourse, gaming, entertainment, technology, workplace experiences, and Gen Z trend spaces.

The current implementation also includes YouTube comments from selected Filipino channels and limited user chat interactions from the deployed system. As of the current live build, the corpus contains 175,481 archived records across Reddit, YouTube, and chat sources. The study does not process private messages, private groups, images, audio, or video content. It also does not claim to provide a complete dictionary of all Filipino slang. Instead, it demonstrates a working system for dynamic slang collection, detection, enrichment, and user interaction.

D. Significance of the Study

This study contributes to Filipino NLP by treating informal language not merely as noisy text but as meaningful linguistic data. By combining corpus collection, burstiness analysis, contextual classification, and LLM-assisted enrichment, *Pinoy Speak* provides a practical framework for studying rapidly evolving Filipino internet slang.

The system also has educational value. Filipino slang often creates an intergenerational and intercultural comprehension gap, particularly when expressions are used in highly contextual online environments. A searchable, contextualized, and continuously updated lexicon can help students, educators, researchers, and general users understand how informal Filipino expressions are used.

As an undergraduate Special Problem, the contribution of this work is application-oriented. The primary output is not only a model or isolated algorithm but a deployed system that integrates data collection, analysis, lexicon management, and user-facing learning tools into one usable web application.

E. Alignment with the UN Sustainable Development Goals

This study aligns with several United Nations Sustainable Development Goals (SDGs).

SDG 4: Quality Education. The system supports language learning and comprehension by providing contextual explanations of informal Filipino expressions.

SDG 9: Industry, Innovation, and Infrastructure. The project demonstrates locally relevant AI and NLP system development for a low-resource and code-switched language environment.

SDG 10: Reduced Inequalities. By focusing on informal Filipino and Taglish rather than only formal English or formal Filipino, the system contributes to more inclusive language technology for Filipino users.

II. Review of Related Literature

A. Code-Switching and Taglish

Code-switching is a common feature of multilingual communities. Poplack [2] describes code-switching as a systematic linguistic practice rather than a random mixture of languages. In the Philippine setting, Tagalog-English code-switching, commonly referred to as Taglish, has been described as a meaningful mode of discourse that reflects social context, identity, and communicative purpose [1], [3].

For NLP systems, Taglish is difficult because it combines tokens, grammatical structures, and discourse cues from more than one language. A sentence may contain English content words, Filipino function words, localized slang, and non-standard spelling in a single utterance. This makes it challenging for tools trained only on formal Filipino or formal English to correctly interpret meaning.

B. Informal Language and Internet Slang

Internet communication encourages rapid lexical innovation. Crystal [7] notes that digital environments create new forms of written expression because users adapt language to speed, identity, humor, and platform conventions. In Filipino online spaces, these conditions produce expressions that may be short-lived, community-specific, or highly dependent on current events.

Slang is especially difficult to model because its meaning is not always recoverable from the word form alone. Some terms are structurally novel, while others are existing words with shifted meanings. For instance, an English word may be borrowed into Filipino online discourse and acquire a new pragmatic meaning. Therefore, detecting slang requires more than checking whether a token appears in a dictionary.

C. Low-Resource Filipino NLP

Filipino NLP remains a relatively low-resource area compared with English and other widely studied languages. The availability of tools such as calamanCy provides an important foundation for Tagalog processing because it offers spaCy-based pipelines for tokenization and other NLP tasks [8]. However, informal Filipino and Taglish still pose additional challenges because social media text often contains code-switching, creative spelling, abbreviations, and emerging vocabulary.

For systems such as *Pinoy Speak*, low-resource NLP tools must be combined with domain-specific heuristics and corpus-based analysis. This is necessary because a general Filipino NLP pipeline may identify tokens but may not automatically determine whether a term is slang, standard vocabulary, profanity, a named entity, or a context-dependent semantic shift.

D. Human Annotation and Validation in Filipino NLP

Human judgment remains important in Filipino NLP tasks that involve context-sensitive meaning. Ocampo and Clariño [9] developed a Filipino metaphor identification study and used manual data annotation by the researcher and four native Filipino annotators. Their annotation team included students with communication and language-related backgrounds, and their process involved checking and normalizing Filipino text before model training. This demonstrates that computational processing alone may be insufficient when the target phenomenon depends on context, figurative use, or local linguistic knowledge.

This is relevant to the present study because Filipino internet slang is also context-sensitive. A term may be ordinary vocabulary in one sentence but slang in another. Some words also require cultural familiarity to determine whether the definition, usage example, and formation type are acceptable. Therefore, this study uses Filipino or Taglish-speaking validators to assess the generated lexicon entries rather than relying only on automatic detection.

E. Subword-Aware Embeddings and OOV Handling

Subword-aware embeddings are useful for handling rare, misspelled, and morphologically varied words. Bojanowski *et al.* [10] introduced FastText word representations that model words as bags of character n-grams. This allows a system to generate useful representations even for words that are rare or unseen during training.

This property is relevant to informal Filipino text because slang often appears in creative or non-standard forms. Character n-gram representations can provide nearest-neighbor evidence for candidate terms, helping the system determine whether a word resembles known slang, standard Filipino morphology, or unrelated noise.

F. Semantic Shift and Diachronic Language Change

Semantic shift refers to changes in word meaning over time or across contexts. Hamilton *et al.* [4] showed that diachronic word embeddings can reveal statistical patterns of semantic change. Kutuzov *et al.* [5] further surveyed methods for tracking semantic shifts using word embeddings, while Bamler and Mandt [6] proposed dynamic word embeddings for modeling language change through time.

These studies support the idea that language change can be observed computationally through distributional patterns. For Filipino slang detection, this is important because some terms are not new words but old words used in new ways. A trend-analysis system must therefore consider both novel out-of-vocabulary words and known words that may be undergoing contextual or semantic change.

G. Retrieval-Augmented Generation

Large Language Models can generate fluent explanations, but they may also hallucinate or provide unsupported definitions when asked about specialized or newly emerging terms. Retrieval-Augmented Generation (RAG) addresses this limitation by combining generation with retrieval from an external knowledge source [11]. Instead of relying only on model memory, a RAG system retrieves relevant documents or entries and uses them as context for the generated response.

This is useful for a slang tutor because the chatbot can ground its explanations in the live lexicon. In *Pinoy Speak*, the tutor module retrieves relevant dictionary entries before producing responses, reducing the risk of inventing unsupported slang meanings.

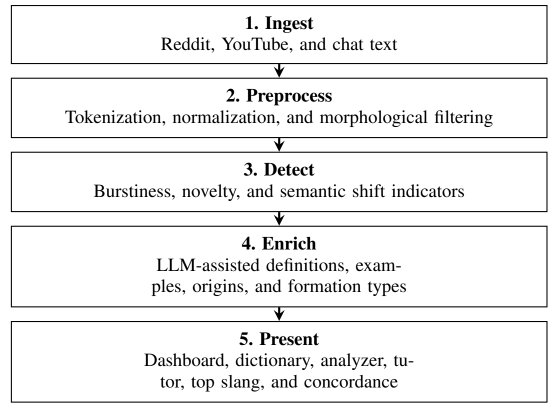

Fig. 1. Operational architecture of Pinoy Speak.

H. Usability Evaluation

System usability is an important part of evaluating applied software systems. Brooke's System Usability Scale (SUS) provides a widely used framework for measuring perceived usability through user responses [12]. Although this study does not implement the full SUS scoring procedure, it follows the same principle of collecting user ratings on navigation, clarity, usefulness, and confidence after system use. This supports the evaluation of *Pinoy Speak* not only as an NLP pipeline but also as a usable educational web application.

III. Materials and Methods

A. System Architecture

The system implements a full-stack modular architecture for continuous informal text collection, slang detection, lexicon enrichment, and user interaction. The backend is built with **FastAPI** and deployed on Fly.io. It manages corpus collection, token processing, slang classification, lexicon storage, LLM enrichment, and chatbot endpoints. The frontend is built using **Next.js 14 App Router** and deployed on Vercel. It provides the dashboard, dictionary, tutor, translator, concordance, and other user-facing modules. The overall workflow is shown in Fig. 1.

B. Implementation Environment

The backend uses Python-based NLP and data-processing components. The system uses calamanCy for Tagalog tokenization and NLP preprocessing [8], the *jcblaise/roberta-tagalog-base* tokenizer as a subword novelty signal, and FastText through *gensim* for word embedding and nearest-neighbor lookup. FastText was selected because character n-gram representations are suitable for rare, misspelled, and morphologically variable words [10]. Corpus records are stored in SQLite, while the verified lexicon is maintained through JSON data files and loaded into the web application.

C. Data Collection

The system collects public text from Reddit posts and comments, YouTube comments, and logged user chat interactions. As of the current implementation, the corpus contains 175,481 archived records: 163,372 from Reddit, 12,075 from YouTube comments, and 34 from chat interactions.

Reddit collection runs every 10 minutes and targets 15 communities selected for high Taglish and slang density: *peyups*, *ADMU*, *dlsu*, *Tomasino*, *RateUPProfs*, *studentsph*, *CasualPH*, *OffMyChestPH*, *mobilelegendsPINAS*, *phgaming*, *OPM*, *PinoyProgrammer*, *ChikaPH*, *TiktokPH*, and *BPOinPH*. These sources represent university life, gaming, entertainment, programming, workplace experiences, and youth-oriented online discourse. YouTube comments are collected hourly from selected Filipino channels through a scraper module.

D. Preprocessing and Morphological Filtering

Informal Filipino text requires preprocessing because it contains code-switching, non-standard spelling, punctuation variation, and highly productive Filipino morphology. The preprocessing stage applies the following steps:

1) **Tokenization:** Input text is tokenized using calamanCy and normalized for further analysis.
2) **Dictionary Gating:** Tokens are checked against standard English and Filipino wordlists. Unknown words are treated as possible novel slang candidates, while known words are monitored for possible semantic shift.
3) **Morphological Prefix Filtering:** Common Filipino affixes and prefix patterns, such as *naka-*, *napaka-*, *nag-*, and *pinaka-*, are filtered to reduce false positives caused by ordinary Tagalog conjugation.

This filtering step was added because early candidate lists contained many standard Tagalog inflected forms that were novel only to the tokenizer but not actually slang.

E. Slang Detection Pipeline

The detection pipeline identifies two main candidate types: novel out-of-vocabulary terms and known words that may be undergoing semantic shift. The simplified procedure is shown in Algorithm 1.

1) Subword Novelty Signal: The system uses a Tagalog RoBERTa tokenizer as a heuristic signal for lexical novelty. If a token is broken into multiple subword units, it is more likely to be unfamiliar or non-standard. This does not automatically classify a word as slang, but it helps prioritize tokens for further checking.

2) Burstiness and Trend Analysis: To detect sudden increases in usage, the system computes a standardized daily frequency score. For a token t, the burstiness score is defined as:

$$Z(t) = \frac{f_{t,daily} - \mu_{t,hist}}{\sigma_{t,hist} + \epsilon} \tag{1}$$

where $f_{t,daily}$ is the token frequency in the current daily window, $\mu_{t,hist}$ is the historical mean frequency, $\sigma_{t,hist}$ is the historical standard deviation, and ϵ prevents division by zero. The implementation uses two thresholds: novel OOV terms are flagged when $Z(t) > 2.0$, while known words monitored for semantic shift are flagged when $Z(t) > 1.8$. This distinction prevents the system from applying the same confidence requirement to all word types.

Algorithm 1 Slang Candidate Detection Pipeline

Require: Corpus C, dictionaries D_{eng}, D_{fil}, tokenizer T_{sub}
Require: Thresholds $\tau_{novel} = 2.0$, $\tau_{shift} = 1.8$
$Posts \leftarrow$ FetchPublicText(C)
$Tokens \leftarrow$ TokenizeAndNormalize($Posts$)
$Candidates \leftarrow \emptyset$
for each token $t \in Tokens$ **do**
  if HasStandardPrefix(t) **then**
    continue
  end if
  $Z(t) \leftarrow$ CalculateDailyZScore(t)
  if $t \in D_{eng} \cup D_{fil}$ **then**
    if $Z(t) > \tau_{shift}$ **then**
      $Candidates$.add($\langle t$, semantic_shift$\rangle$)
    end if
  else
    $subwords \leftarrow T_{sub}(t)$
    if $|subwords| \geq 2$ **and** $Z(t) > \tau_{novel}$ **then**
      $type \leftarrow$ ClassifyFormation(t)
      $Candidates$.add($\langle t, type\rangle$)
    end if
  end if
end for
return $Candidates$

3) Formation Type Classification: Candidates that pass the novelty and burstiness checks are categorized according to structural formation. The classifier identifies types such as *binaliktad* or syllable reversal, *jejemon* or alphanumeric spelling, number-syllable forms, semantic shifts, and unknown formations.

F. Context-Window Gate and Classification Precedence

Some seed entries are marked with an `is_ambiguous` flag because they may occur either as ordinary English or Filipino words or as slang. Examples include *bet*, *solid*, *ghost*, *slay*, *sus*, *chika*, and *tropa*. To reduce false positives, the `/analyze` endpoint checks surrounding tokens for Filipino grammatical particles and intensifiers such as *ang*, *ng*, *sa*, *naman*, *kasi*, *talaga*, *sobrang*, and *grabe*. For example, *bet* in "I bet" is not confirmed as slang, while *bet* in a Filipino-context sentence may be accepted.

The classifier also applies slang-before-profanity precedence. The slang lexicon is checked before the profanity list so that culturally meaningful words that appear in both resources can still be represented with context rather than being automatically suppressed.

G. LLM-Based Slang Enrichment

Verified candidates are passed to a separate enrichment pipeline. This pipeline uses Google Gemini as the primary provider, Groq Llama 3.1 8B as fallback, and a local Ollama model as a last-resort provider. The enrichment process generates a structured dictionary entry containing a definition, plain-language explanation, part of speech, possible origin, example usage, and formation type. Accepted entries are stored in `discovered_slang.json` and merged with the seed lexicon.

H. Chatbot LLM and Retrieval-Augmented Lexicon Access

The user-facing "Kuya Slang" tutor is separate from the enrichment pipeline. It is accessed through the frontend `/tutor` route. The chatbot uses Groq Llama 3.3 70B as the primary model with Gemini 2.0 Flash as fallback. To reduce hallucination, the backend builds a vector index from the live lexicon at startup through `rag_store.py`. When the user asks an open-ended question, the tutor retrieves the nearest dictionary entries and uses them as grounding context. This follows the principle of retrieval-augmented generation, where external knowledge is retrieved before producing a response [11].

I. User-Facing Features

The deployed frontend contains the following routes:

- `/` – live statistics dashboard with corpus metrics and trend charts;
- `/dictionary` – searchable and filterable slang lexicon;
- `/tutor` – Kuya Slang tutoring chatbot;
- `/translator` – Taglish-to-English sentence translator;
- `/top-slang` – top slang leaderboard;
- `/concordance` – in-context corpus search; and
- `/about` – creator and project information page.

The backend also includes administrative endpoints for lexicon growth. The `/sweep-corpus` endpoint scans existing corpus records for high-frequency OOV tokens and sends them through the detection pipeline. The `/import-online-slang` endpoint imports slang candidates from external Filipino slang sources and Reddit search results, then verifies and enriches accepted entries.

J. Ethical Considerations

The study uses only publicly accessible text sources and excludes private groups, private videos, private messages, images, and audio. Usernames and identifying metadata are not used in the displayed dictionary entries. Example sentences may be paraphrased to reduce the possibility of tracing text back to individual users. The frontend also includes a toggleable profanity filter to support safer browsing of sensitive lexical content.

K. Evaluation Procedure

The deployed application was evaluated using two complementary instruments: a lexicon validation form and a web application usability and bug tracking form. The evaluation was designed as a beta evaluation of the implemented *Pinoy Speak* web application, where selected users and validators interacted with the deployed system and provided feedback on both lexicon quality and application usability.

The use of human validators was adapted from prior Filipino NLP work that employed native Filipino annotators for manual language annotation. Ocampo and Clariño [9] used the researcher and four native Filipino annotators to manually annotate Filipino text, showing that human judgment remains necessary for Filipino NLP tasks involving context-sensitive meaning. In the present study, validators were not used to train a classification model but to evaluate the quality of the generated slang lexicon.

For lexicon validation, 79 dictionary entries were divided into three form parts to reduce annotator fatigue. Each lexical item was evaluated by three Filipino or Taglish-speaking validators. Across the three form parts, four unique validators participated, but every lexical item received three item-level judgments. For each word, validators assessed whether the term is actually used as Filipino or Taglish slang, whether the system definition is correct, whether the example sentence sounds natural, whether the assigned formation type is acceptable, and whether the final entry is acceptable, needs minor revision, or should be treated as an incorrect or false-positive entry. Definition correctness and example naturalness were rated on a five-point scale. Final judgments and written comments were used to identify entries requiring dictionary revision.

For usability evaluation, a separate beta testing survey asked respondents to test the deployed web application and evaluate six modules: live dashboard or word popularity, navigation flow, top slang or usage frequency, dictionary or word cards, analyzer or translator, and tutor or chatbot. The form collected module-level usability ratings, five-point ratings for interface aesthetics, loading speed, device compatibility, clarity of instructions, and data accuracy, as well as bug occurrence and severity reports. The usability dataset contained 39 submitted responses. After excluding two non-consenting responses and one blank-consent response, 36 consented responses were retained for analysis.

To make the beta evaluation more concrete, the survey included task-based testing examples that required users to interact with the system as actual end users. One beta testing example asked the user to open the Analyzer or Translator module and enter a Taglish sentence containing informal expressions, such as "Bet ko itong outfit mo, sobrang solid." The expected behavior was for the system to recognize context-dependent slang terms such as *bet* and *solid*, explain their informal meaning, and avoid treating them only as ordinary English words. This task was included because the same word may have different meanings depending on the surrounding context. For example, *bet* in "I bet you are right" is an ordinary English verb, while *bet* in a Filipino or Taglish sentence may express approval, preference, or liking.

Another beta testing task asked users to check the Dictionary module by searching for selected slang entries and reviewing whether the displayed definition, example sentence, and formation type were understandable. Users were also asked to test the Tutor or Chatbot module by asking a slang-related question and evaluating whether the answer was clear, helpful, and consistent with the lexicon. These task-based examples allowed the evaluation to assess not only the visual interface but also whether the application supported its intended purpose as a contextual slang-learning system.

Quantitative responses were summarized using means and percentages, while open-ended responses were grouped into recurring themes such as vocabulary expansion, accuracy improvement, better examples, pronunciation support, regional coverage, and interface refinements.

IV. Results and Discussion

A. System Implementation and Live Lexicon

The implemented *Pinoy Speak* system is currently deployed as a full-stack web application with a FastAPI backend and a Next.js frontend. The system collects public text from Reddit, YouTube comments, and user chat interactions. As of the current implementation, the corpus contains 175,481 archived records: 163,372 Reddit records, 12,075 YouTube comments, and 34 chat interaction records.

The live lexicon contains approximately 104 verified slang entries. This includes 102 seed entries and dynamically discovered entries. Among the seed entries, 29 are marked as ambiguous because they require surrounding context before being classified as slang. These include words such as *bet*, *solid*, *ghost*, *slay*, *sus*, *chika*, and *tropa*. The formal lexicon validation, however, covered 79 selected entries with three ratings per entry. This distinction is important because the live system contains both seed and dynamically discovered entries, while the validation set represents the subset formally reviewed by human validators.

The resulting system demonstrates the feasibility of maintaining a dynamic lexicon that combines seeded knowledge, corpus-based discovery, and LLM-assisted enrichment. The deployed application also integrates the core modules needed for slang exploration and learning, including the live dashboard, dictionary, tutor, translator, top slang leaderboard, concordance search, and about page. The frontend is hosted on Vercel, while the backend is hosted on Fly.io, allowing users to access the system through a browser without installing local software.

B. Observed Reduction of False Positives

Early system outputs showed that standard Tagalog inflected words could be incorrectly flagged as novel slang because they were unfamiliar to subword or dictionary checks. To address this, the system added a morphological prefix filter for common Filipino affix patterns such as *naka-*, *napaka-*, *nag-*, and *pinaka-*. This filtering step reduced observed false positives in candidate lists by excluding common standard Tagalog conjugations before novelty and burstiness checks.

A second source of false positives came from ambiguous English or Filipino words that can function as slang only in certain contexts. The context-window gate addresses this by checking nearby Filipino particles and intensifiers before confirming ambiguous entries. This allows the system to reject English-context uses such as “I bet” while still accepting Filipino-context uses of terms such as *bet*, *solid*, and *grabe*. These controls improve the interpretability of the resulting lexicon, although a full before-and-after quantitative false-positive analysis remains part of future work.

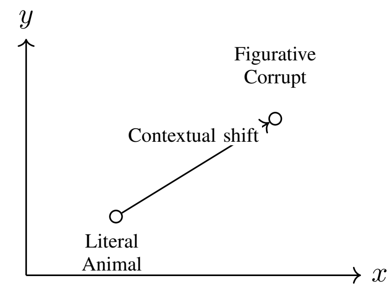

Fig. 2. Conceptual illustration of semantic shift from literal to figurative usage.

C. Detection of Semantic Shifts

The system can identify known words that may be undergoing contextual or semantic shift. Instead of treating all dictionary words as non-candidates, the pipeline monitors known words for unusual increases in frequency. If a known term exceeds the semantic-shift threshold, it is flagged for review and enrichment.

Fig. 2 presents a conceptual illustration of this process using the Filipino word *buwaya*. In its literal meaning, *buwaya* refers to a crocodile. In informal political discourse, however, it may be used figuratively to refer to corruption or greed. The figure is not presented as a direct output of a measured embedding experiment; rather, it illustrates how vector-space movement can represent a shift from literal to figurative contexts.

D. Interactive Chatbot and RAG-Grounded Explanations

The “Kuya Slang” tutor, accessed through the `/tutor` route, provides conversational explanations of slang terms. Unlike the enrichment pipeline, which is used for generating dictionary entries, the tutor uses a separate chatbot LLM stack and retrieves relevant entries from the live lexicon before answering. This retrieval step helps ground responses in verified dictionary content and reduces the risk of unsupported explanations.

The tutor is particularly useful for learners who need more than a one-word definition. It can explain usage, give examples, and clarify the difference between literal and slang meanings. This supports the educational goal of the system by turning the lexicon into an interactive learning resource rather than a static list.

E. Lexicon Validation Results

A separate lexicon validation was conducted to determine whether the system-generated dictionary entries were acceptable to Filipino or Taglish speakers. The validation covered 79 slang entries across three form parts, with each entry receiving three item-level ratings. This produced 237 total item-level judgments. Although four unique validators participated across the form parts, each lexical item was still evaluated by three validators.

Overall results were positive. Validators marked 227 of 237 item-level judgments (95.78%) as actually used in Filipino or Taglish slang. Definition correctness obtained a mean score of

TABLE I
SUMMARY OF LEXICON VALIDATION RESULTS

Validation Metric	Result
Validated slang entries	79
Total item-level ratings	237
Actually used as slang	95.78%
Definition correctness	4.83 / 5
Example naturalness	4.75 / 5
Formation type acceptable	93.67%
Final judgment: correct/acceptable	87.76%
Final judgment: needs minor revision	9.70%
Final judgment: incorrect/false positive	2.53%

4.83 out of 5, while example sentence naturalness obtained a mean score of 4.75 out of 5. Formation type acceptability was marked "Yes" in 222 of 237 judgments (93.67%). Final entry judgments were also favorable: 208 of 237 judgments (87.76%) were marked as correct or acceptable, 23 (9.70%) needed minor revision, and 6 (2.53%) were marked as incorrect or possible false positives.

At the word level, 55 of 79 entries received unanimous correct or acceptable judgments from all three validators. Twenty entries received two correct or acceptable judgments and one revision or rejection judgment. Four entries received one or fewer correct judgments: *basic*, *dead*, *dogshow*, and *yarn*. These entries require priority revision because validator comments indicated possible definition mismatch or insufficient contextual explanation. For instance, validators noted that *dogshow* is more accurately understood as playful teasing, mocking, or making fun of someone, while *yarn* should be explained as a stylized form of *iyan* or *yan* rather than treated as a completely independent lexical origin.

Validator comments also showed that some terms are acceptable as entries but require metadata corrections. Examples include *bonak*, whose formation was described as a contraction related to *bobong anak*; *omsim* and *lodi*, which should be consistently labeled as reversed forms; and borrowed English items such as *legit*, *savage*, *shook*, and *shookt*, whose formation labels should more clearly identify borrowing or semantic shift. These comments are important because they show that system accuracy is not limited to meaning alone; origin, formation type, and example naturalness also affect lexicon quality.

F. User Evaluation Results

A usability evaluation survey was conducted as a beta evaluation to assess the usability, clarity, perceived usefulness, and bug behavior of the deployed *Pinoy Speak* web application. The survey received 39 submissions. After excluding two non-consenting responses and one blank-consent response, 36 valid responses were retained. Respondents came from several age groups, with 18–24 as the largest group (17 respondents), followed by 25–34 (9 respondents), 45 and above (4 respondents), under 18 (4 respondents), and 35–44 (2 respondents). The respondent group included 18 students, 15 employed users, and three respondents from other categories. Windows was the most frequently used operating system, while Chrome was the most frequently used browser.

TABLE II
MODULE-LEVEL USABILITY AND QUALITY RATINGS

Module / Task	Use.	Aes.	Spd.	Acc.
Dashboard / Word Popularity	8.58	4.33	4.47	4.25
Navigation / Menu Flow	9.14	4.53	4.44	4.42
Top Slang / Usage Frequency	8.85	4.50	4.56	4.33
Dictionary / Word Cards	9.36	4.47	4.50	4.33
Analyzer / Translator	9.00	4.64	4.53	4.42
Tutor / Chatbot	9.33	4.58	4.61	4.44

Note. Use. = usability; Aes. = aesthetics; Spd. = speed; Acc. = data accuracy.

TABLE III
REPORTED BUG OCCURRENCE BY MODULE

Module / Task	Reports	Percent
Dashboard / Word Popularity	6	16.67%
Navigation / Menu Flow	1	2.78%
Top Slang / Usage Frequency	4	11.11%
Dictionary / Word Cards	1	2.78%
Analyzer / Translator	1	2.78%
Tutor / Chatbot	1	2.78%

The beta evaluation used task-based testing to check whether users could interact with the major modules of the system. One example task involved the Analyzer or Translator module, where users could enter a Taglish sentence such as "Bet ko itong outfit mo, sobrang solid." This example tested whether the system could interpret ambiguous slang expressions using context. In this case, *bet* should be interpreted as approval or preference in a Filipino or Taglish sentence, while *solid* may function as an informal positive description. The expected output was that the Analyzer or Translator would identify these terms as informal Taglish expressions based on surrounding Filipino context words such as *ko*, *itong*, and *mo*. This example shows that the beta evaluation did not only measure interface usability but also tested whether the system could support contextual slang understanding.

Across the evaluated modules, the application received high usability scores. As shown in Table II, the Dictionary module obtained the highest average overall usability score at 9.36 out of 10, followed by the Tutor module at 9.33, Navigation Flow at 9.14, Analyzer or Translator at 9.00, Top Slang or Usage Frequency at 8.85, and Dashboard or Word Popularity at 8.58. These results suggest that users found the core lexicon and learning-oriented features especially usable. The Analyzer or Translator score of 9.00 out of 10 also supports the usefulness of including concrete beta testing tasks involving contextual slang interpretation.

Bug reports were generally low, but they identify practical areas for refinement. The Dashboard or Word Popularity module received the highest number of bug reports at 6 of 36 responses (16.67%), followed by the Top Slang or Usage Frequency module with 4 reports (11.11%). The remaining evaluated modules each received only one bug report. Reported issues included unclear zooming in charts, delayed or missing chart rendering, search behavior in the dictionary, minor animation shifts in the analyzer, and occasional tutor-answer mismatches.

At the general application level, respondents gave strong

TABLE IV
General Application-Level Evaluation Results

Evaluation Item	Mean
Interface was visually clear and readable	4.72 / 5
App was easy to navigate	4.67 / 5
Dictionary was useful	4.61 / 5
Would recommend Pinoy Speak	4.61 / 5
Helped understand slang context	4.50 / 5
Tutor helped users understand slang	4.42 / 5
Definitions/examples felt accurate and natural	4.25 / 5
Improved confidence in understanding slang	4.25 / 5
Compared favorably with other slang-learning sources	3.94 / 5

ratings for readability, navigation, usefulness, and recommendation. The interface was rated visually clear and readable at 4.72 out of 5, while ease of navigation was rated 4.67 out of 5. The Dictionary was rated 4.61 out of 5 for usefulness, and the recommendation item also received 4.61 out of 5. The application helped users understand the context of slang, not just its meaning, with a mean rating of 4.50 out of 5. In addition, 31 of 36 respondents (86.11%) reported that they learned at least one new slang word or expression from the application.

The most frequently selected useful features were the Dictionary and the overall feature set, with 12 selections each. Other useful features included Concordance, Chat Tutor, Translator, and Top Slang. Qualitative feedback emphasized the need to expand the slang vocabulary, improve selected definitions and translations, add pronunciation support, include regional Philippine languages or slang varieties, freeze or improve dictionary navigation controls, clarify chart interactions, and separate quiz or flashcard activities from the tutor interface.

Overall, the beta evaluation results indicate that the deployed application is usable for its intended users and that the main learning-oriented modules are understandable. However, the feedback also shows that the application still requires iterative improvement, especially in expanding the slang database, refining selected definitions and translations, improving chart interactions, and reducing occasional tutor-answer mismatches.

G. Discussion

The results show that *Pinoy Speak* addresses the main problem identified in this study: informal Filipino and Taglish evolve faster than static dictionaries and conventional NLP tools can accommodate. The implemented system responds to this problem by combining corpus-based trend detection, contextual classification, LLM-assisted enrichment, and user-facing educational modules.

The lexicon validation results support the usefulness of the dictionary, but they also show that a dynamic lexicon requires human review. High definition and example scores indicate that most entries are understandable and natural. However, the minor revision and false-positive judgments show that some entries need stronger distinction between slang, ordinary Filipino vocabulary, borrowed English usage, and context-dependent semantic shift. This is especially important for entries such as *bawi*, *malas*, *swerte*, *tampo*, and *wagi*, where validators questioned whether the word is slang or standard vocabulary.

The usability and beta evaluation results show that users value the Dictionary, Tutor, and Analyzer modules. These modules directly support the educational aim of the project because they help users look up meanings, see examples, and ask follow-up questions. The beta testing example using a Taglish sentence such as "Bet ko itong outfit mo, sobrang solid" also demonstrates why context-aware analysis is necessary. The system must not only detect individual slang terms but also interpret them based on the surrounding words. However, the lower ratings and higher bug reports for chart-oriented modules suggest that trend visualizations should be refined further. Users need clearer chart instructions, more visible loading states, and more intuitive interactions when inspecting frequency data.

Overall, the combined evaluation shows that the system is usable and promising, but not yet a fully automatic authority on Filipino slang. Its strongest role is as a semi-automated dynamic lexicon system: it collects and prioritizes candidate terms computationally, then improves reliability through validation, correction, and iterative updating.

H. Limitations and Future Work

Although the system is deployed and functional, several limitations remain. First, the corpus is drawn from selected public online communities and therefore does not represent all Filipino speakers, regions, or age groups. Second, the usability respondents were mostly young adult users and students, so the usability results should be interpreted as initial user feedback rather than a population-wide evaluation. Third, the lexicon validation used three ratings per term, which is useful for initial validation but still limited for measuring broad agreement across dialects, regions, age groups, and social communities.

Future work should expand the corpus to include more regional Filipino languages and dialectal varieties, improve human validation of discovered slang entries, add audio pronunciation support, strengthen the community submission workflow, and formally measure false-positive and false-negative rates. The immediate next revision of the lexicon should prioritize entries with low final acceptability or strong validator comments, especially *dogshow*, *yarn*, *basic*, and *dead*, followed by metadata corrections for formation type and origin.

V. Conclusion and Recommendations

A. Conclusion

This study developed and evaluated *Pinoy Speak*, a deployed full-stack system for tracking, organizing, and explaining informal Filipino and Taglish slang. The system addresses the limitations of static dictionaries by continuously collecting public online text, identifying candidate slang terms, enriching verified entries, and presenting them through user-facing learning tools.

The current implementation archives 175,481 records from Reddit, YouTube comments, and user chat interactions. It maintains approximately 104 verified slang entries composed of seed entries and dynamically discovered terms. The system uses a multi-stage NLP pipeline that combines calamanCy tokenization, dictionary gating, subword novelty signals, FastText-based similarity support, temporal burstiness analysis, morphological filtering, and context-window disambiguation. These components allow the system to detect both novel out-of-vocabulary slang and known words that may be undergoing semantic shift.

The evaluation shows that the system is promising but still requires iterative human validation. The 79-entry lexicon validation produced strong ratings for definition correctness (4.83/5) and example naturalness (4.75/5). Most item-level judgments were marked correct or acceptable, but validators also identified entries that need revision, especially cases where the word may be ordinary Filipino vocabulary, borrowed English, or a context-dependent expression rather than clear slang. This confirms that human review remains necessary for maintaining a reliable living dictionary.

The 36-response consented beta usability evaluation showed positive user acceptance. The Dictionary received the highest module usability rating at 9.36 out of 10, followed by the Tutor at 9.33, Navigation Flow at 9.14, Analyzer or Translator at 9.00, Top Slang at 8.85, and Dashboard or Word Popularity at 8.58. General ratings also showed that respondents found the interface visually clear, easy to navigate, useful, and recommendable. In addition, 86.11% of respondents reported learning at least one new slang word or expression from the system.

Overall, the study shows that informal Filipino and Taglish slang can be modeled as dynamic linguistic data rather than treated merely as noise. By integrating corpus-based trend analysis, contextual disambiguation, LLM-assisted enrichment, human lexicon validation, and an educational web interface, *Pinoy Speak* provides an application-oriented contribution to Filipino NLP and digital language learning.

B. Recommendations

Although the implemented system achieved its main objectives, several improvements are recommended for future work.

First, the corpus should be expanded to include more regional Filipino languages, dialectal varieties, and platform sources. The current corpus is useful for detecting slang from selected high-activity online communities, but it does not represent all Filipino speakers or all regional forms of informal language. Future versions should include more balanced data from different regions, age groups, and online communities.

Second, lexicon validation should be expanded. The current validation provided three ratings per term, which is useful for initial checking, but future versions should include more annotators and clearer annotation guidelines. Priority should be given to entries flagged by validators, including terms with unclear slang status, incorrect definitions, weak examples, or questionable formation labels.

Third, the system should improve its handling of ambiguous and borrowed words. Validator comments showed that terms such as *basic*, *dead*, *legit*, *savage*, *shook*, and *shookt* require stronger explanation of semantic shift, borrowing, and local usage. The lexicon should therefore distinguish between ordinary words, borrowed slang, shifted meanings, and community-specific expressions.

Fourth, the user interface should refine chart interactions, dictionary navigation, and tutor activities. Usability feedback suggested clearer graph controls, better loading indicators, a sticky dictionary search bar or alphabet filter, more natural tutor responses, and separate quiz or flashcard features. These refinements would improve both user experience and educational value.

Finally, future evaluation should include more formal computational metrics such as precision, recall, false-positive rate, false-negative rate, and inter-annotator agreement. These measures would strengthen the evidence for system accuracy beyond usability ratings and initial human validation.

C. Final Remarks

Pinoy Speak demonstrates that Filipino internet slang can be documented through a combination of computational detection, human validation, and interactive presentation. The system is not intended to replace expert linguistic judgment. Rather, it serves as a practical tool for tracking emerging expressions, helping users understand context, and supporting future work in low-resource Filipino NLP.

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